Experiment 2

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Aim: To implement Single Page Application (SPA) navigation in the EcoTrack application using React Router, secure application routes using context-based authentication, and extend nested dashboard routing through follow-up enhancements.

Objective:

After completing this experiment and its follow-up tasks, the student will be able to:

1. Configure client-side routing in a React application using React Router
2. Implement SPA navigation without full page reloads
3. Design and apply protected routes using route-guard patterns
4. Manage shared authentication state using React Context API
5. Implement nested routing to build dashboard-style layouts
6. Extend existing nested routes by adding new dashboard sections
7. Implement logout functionality by updating shared context state
8. Analyze route access behavior and explain redirection logic
9. Understand the role of Context API in shared state management and its comparison with Redux at an introductory level.

Implementation/Code:

Header.js:

```
import { Link } from "react-router-dom";
import { useAuth } from "../context/AuthContext";

const Header = () => {   const {
isAuthenticated } = useAuth();

  return (
```

```
    <header style={styles.header}>
      <div style={styles.logo}>  EcoTrack</div>

      <nav style={styles.nav}>
        {isAuthenticated && (
          <>
            <Link to="/">Dashboard</Link>
            <Link to="/logs">Logs</Link>
          </>
        )}
        {!isAuthenticated && <Link to="/login">Login</Link>}
      </nav>
    </header>
  );
};

export default Header;

const styles = {   header: {     height:
"64px",     backgroundColor: "var(--
surface)",
    display: "flex",     alignItems:
"center",     justifyContent: "space-
between",     padding: "0 24px",
    borderBottom: "1px solid #1f2937"
  },   logo: {
fontSize: "20px",
fontWeight: "700",
    color: "var(--primary)"
  },   nav: {
display: "flex",
    gap: "20px"
  }
};
```

AuthContext.js:

```
import { createContext, useContext, useState } from "react";

const AuthContext = createContext(null);
```

```
export const AuthProvider = ({ children }) => {
  const [isAuthenticated, setIsAuthenticated] = useState(false);

  const login = () => {
    setIsAuthenticated(true);
  };

  const logout = () => {
    setIsAuthenticated(false);
  };

  return (
    <AuthContext.Provider
value={{
isAuthenticated,
        login,
logout
      }}
    >
      {children}
    </AuthContext.Provider>
  );
};

export const useAuth = () => useContext(AuthContext);
```

Dashboard.js:

```
const DashboardSummary = () => (
  <>
    <h2>Carbon Summary</h2>
    <p style={{ color: "var(--muted)" }}>
      Overview of your environmental impact.
    </p>
  </>
);

export default DashboardSummary;
```

DashboardAnalytics.js:

```
const DashboardAnalytics =()=>{
```

```
    return (
      <p>This is Dashboard Analytics</p>
    )
  };
  export default DashboardAnalytics;
```

DashboardSummary.js:

```
const DashboardSettings = () => {
  return (
    <h3 style={{color: "skyblue"}}>These are the settings</h3>
  )
}

export default DashboardSettings;
```

DashboardLayout.js:

```
import { Link, Outlet } from "react-router-dom"; import
Logout from "./Logout";

const DashboardLayout = () => {
return (
  <div style={styles.container}>
    <aside style={styles.sidebar}>
      <h3 style={styles.sideTitle}>Dashboard</h3>
      <Link to="summary">Summary</Link>
      <Link to="analytics">Analytics</Link>
      <Link to="settings">Settings</Link>
    </aside>

    <main style={styles.main}>
      <Outlet />
      <Logout />
    </main>
  </div>
 );
};

export default DashboardLayout;

const styles = {
container: {
```

```
    display: "grid",
    gridTemplateColumns: "220px 1fr",
    minHeight: "calc(100vh - 64px)"
  },
  sidebar: {     backgroundColor:
"var(--surface)",     padding: "20px",
display: "flex",     flexDirection:
"column",     gap: "12px",
borderRight: "1px solid #1f2937"
  },   sideTitle: {
color: "var(--accent)",
    marginBottom: "10px"
  },
  main: {
padding: "32px"
  }
}; Login.js:

import { useAuth } from "../context/AuthContext"; import
{ useNavigate } from "react-router-dom";

const Login = () => {   const {
login } = useAuth();   const
navigate = useNavigate();

  const handleLogin = () => {
login();     navigate("/");
  };

  return (
    <div style={styles.wrapper}>
      <div style={styles.card}>
        <h2>Welcome to EcoTrack</h2>
<p style={{ color: "var(--muted)" }}>
          Track and reduce your carbon footprint.
        </p>
        <button onClick={handleLogin}>Login</button>
      </div>
    </div>
  );
};
```

```
export default Login;

const styles = {   wrapper: {
minHeight: "calc(100vh - 64px)",
display: "flex",     alignItems:
"center",     justifyContent: "center"
  },   card: {     backgroundColor:
"var(--surface)",     padding: "32px",
borderRadius: "14px",     width:
"320px",     textAlign: "center"
  }
};
```

Logout.js:

```
import { useAuth } from "../context/AuthContext"; import
{ useNavigate } from "react-router-dom";

const Logout = () => {   const {
logout } = useAuth();   const
navigate = useNavigate();

  const handleLogout = () => {
logout();                    navigate("/login",
{ replace: true });
  };

  return (
    <button onClick={handleLogout}>
      Logout
    </button>
  );
};

export default Logout;
```

ViewLogs.js:

```
import logs from "../data/logs";

const Viewlogs = () => {
    return (
        <div>
            <h2>Experiment Logs</h2>
```

```
                <ul>
                    {logs.map((log, index) => (
                        <li key={log.id}>
                        {log.activity}: {log.carbon} kg CO2
                        </li>
                    ))}
                </ul>
            </div>
        ) } export default
    Viewlogs;
```

PrototectedRoute.js:

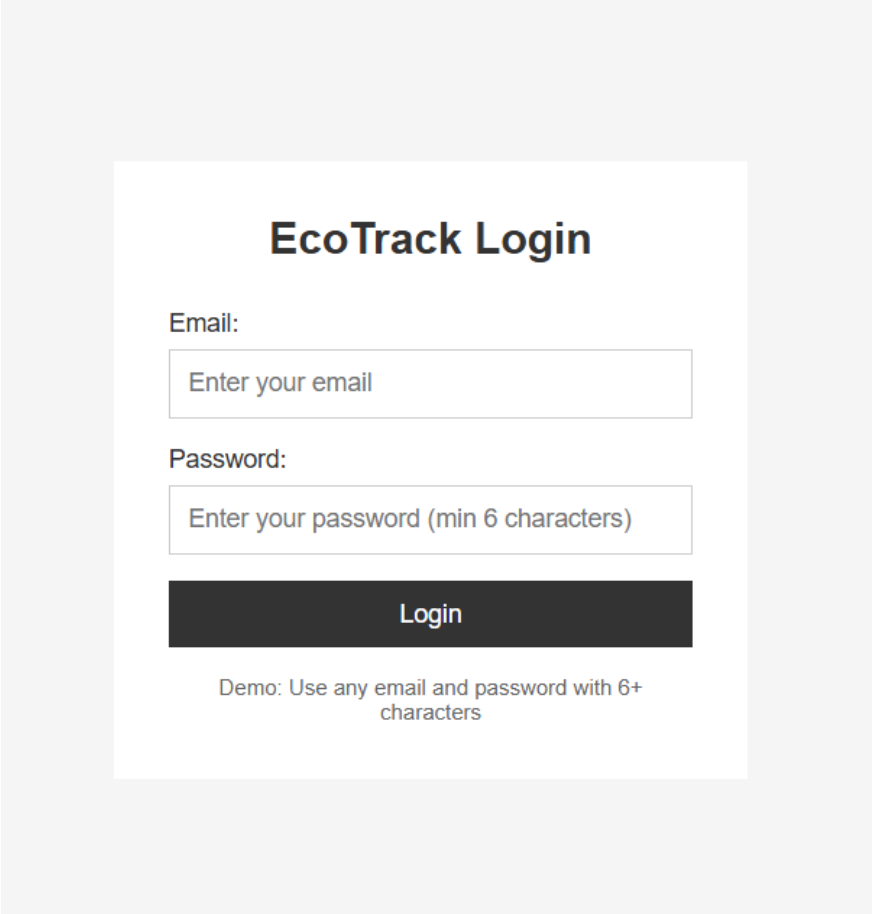
```
import { Navigate } from "react-router-dom";
import { useAuth } from "../context/AuthContext";

const ProtectedRoute = ({ children }) => {   const { isAuthenticated
} = useAuth();   return isAuthenticated ? children : <Navigate
to="/login" replace />; };

export default ProtectedRoute;
```

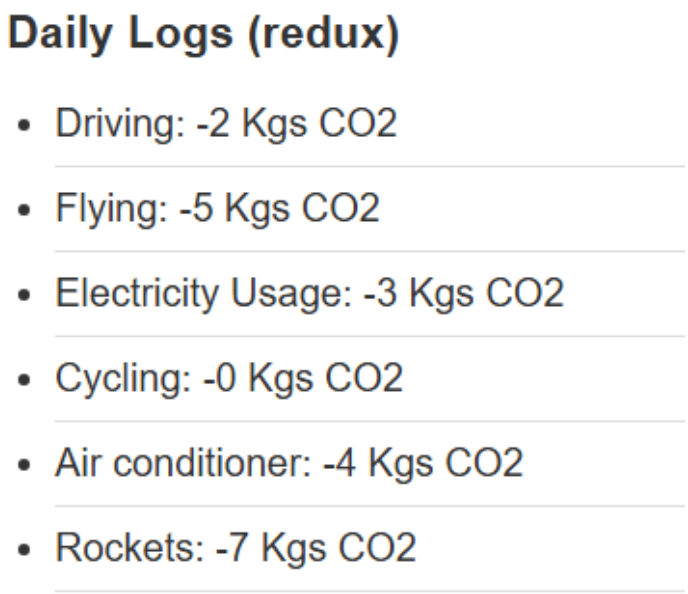
App.js:

```
import { Routes, Route } from "react-router-dom";

import Header from "./components/Header"; import
Login from "./pages/Login";
import Logs from "./pages/logs";

import DashboardLayout from "./pages/DashboardLayout"; import
DashboardSummary from "./pages/DashboardSummary"; import
DashboardSettings from "./pages/DashboardSettings"; import
DashboardAnalytics from "./pages/DashboardAnalytics";

import ProtectedRoute from "./routes/ProtectedRoute";

function App() {
return (
   <>
     <Header />
```

```
    <Routes>
      <Route path="/login" element={<Login />} />

      <Route
path="/"
element={
          <ProtectedRoute>
            <DashboardLayout />
          </ProtectedRoute>
        }
      >
        <Route index element={<DashboardSummary />} />
        <Route path="summary" element={<DashboardSummary />} />
        <Route path="settings" element={<DashboardSettings />} />
        <Route path="analytics" element={<DashboardAnalytics />} />
      </Route>

      <Route
path="/logs"
element={
          <ProtectedRoute>
            <Logs />
          </ProtectedRoute>
        }
      />
    </Routes>
  </>
  );
}

export default App;
```

Output:





Learning Outcome :

- Learnt about react files.
- Learnt Design and apply protected routes using route-guard patterns.
- Learnt Implement nested routing to build dashboard-style layouts.